MARDAS MD2PDF
MARKDOWN TO PDF ENGINE

Complete Guide

Mardas MD2PDF Guide

Complete user manual and feature reference

A complete guide for installing, configuring, and using Mardas MD2PDF.
This document also acts as a live rendering sample for cover pages, tables of contents, mixed RTL/LTR text, formulas, code, Mermaid flowcharts, images, tables, footnotes, page breaks, and safe HTML.

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Introduction

> **Note**
>
> This guide is the complete user manual and feature reference for Mardas MD2PDF. It teaches each supported feature with runnable Markdown and keeps those same examples in the official PDF output.

Mardas MD2PDF is a Markdown-to-PDF publishing tool designed for documents that mix Persian and English content. It keeps the writing workflow simple while giving the final PDF a professional printed layout.

The project is useful for reports, university documents, technical guides, educational notes, research drafts, project documentation, and any Markdown file that needs a clean PDF output.

```text
Markdown -> Structured HTML -> Chromium PDF
```

The renderer does not draw paragraphs manually on a PDF canvas. Instead, it converts Markdown into a structured HTML document, applies print-oriented CSS, renders formulas with MathJax, and asks Chromium to produce the final PDF. This gives the project better support for CSS print layout, mixed text direction, SVG formulas, syntax-highlighted code, local images, and complex tables.

> **Note**
>
> This guide is both documentation and a rendering sample. The PDF version of this file is available in the examples/ directory, so users can inspect the actual output of every major feature.

> **Tip**
>
> Mardas MD2PDF now keeps user-facing feature documentation in this guide instead of maintaining parallel feature-reference pages. Release, maintenance, security, and changelog files remain under docs/ for project operations.

Rendering sample checklist

The guide intentionally contains compact test cases for the renderer. When you review the generated PDF, check that these samples appear correctly:

Captioned images, tables, code listings, and Mermaid diagrams are now normalized as semantic print blocks so captions remain attached to their associated content in the generated PDF.

Sample area	What to verify in the PDF
Cover and metadata	Title, subtitle, authors, summary, version, status, keywords, and language-specific labels.
TOC and outline	Nested heading numbers, internal links, and PDF viewer bookmarks generated from Markdown headings.
Mixed direction text	Persian/English text, inline code, and identifiers remain readable in the same paragraph.
MathJax	Inline math aligns with text and display equations are centered and scaled.
Code blocks	Fenced, indented, and language-less code blocks render without corrupting content.
Mermaid	A `flowchart` code fence becomes an SVG diagram instead of a plain code block.
Images and HTML	Local Markdown images and safe HTML image tags appear in the PDF.
Footnotes and page flow	Numeric footnote references, repeated-reference back-links, multiline footnotes, manual page breaks, margins, and footer numbering remain stable.
Persian/RTL audit	Persian punctuation, Latin/Persian numerals, RTL tables, mixed-script TOC headings, captions, and footnote backlinks remain readable.

Main capabilities

Mardas MD2PDF focuses on the features that matter most for polished technical PDFs:

Capability	Description
Persian and English documents	`lang: fa`, `lang: en`, RTL/LTR shell direction, and mixed inline text.
Cover pages	Title, subtitle, authors, summary, logo, date, version, status, keywords, and academic metadata.
Table of contents and outline	Hierarchical TOC plus PDF viewer bookmarks generated from Markdown headings.
MathJax	Inline and display formulas with browser-rendered output.
Code blocks	Pygments syntax highlighting for fenced and indented code blocks.
Mermaid flowcharts	Offline SVG rendering for practical `flowchart` / `graph` diagrams.
Local images	Markdown and safe HTML images can be embedded as data URIs.
Safe HTML	Raw HTML is sanitized by default.
Footnotes	Multiline Markdown footnotes are supported.
Appearance system	Separate `style`, `palette`, and `mode` choices for layout shape, color, and light/dark output.
Automation	CLI workflow suitable for local scripts and CI jobs.
GUI	Local browser-based editor, preview, option selector, and exporter.

Installation

Requirements

Before using the project, make sure you have:

- Python 3.10 or newer;
- a working virtual environment;
- Playwright Chromium installed;
- a Persian-capable font on the system, preferably Vazirmatn;
- Git, if you plan to clone the repository.

Install from source

```bash
git clone https://github.com/mragetsars/Mardas-MD2PDF.git
cd Mardas-MD2PDF
python -m venv .venv
source .venv/bin/activate
pip install -e .
python -m playwright install chromium
```

On Windows PowerShell:

```powershell
python -m venv .venv
.venv\Scripts\Activate.ps1
pip install -e .
python -m playwright install chromium
```

Development installation

For development, install the optional test dependencies:

```bash
pip install -e .[dev]
pytest
```

The package exposes three process interfaces:

Command	Purpose
`mrs-md2pdf`	Convert Markdown files to PDF from the command line.
`mrs-md2pdf-gui`	Launch the current local browser-based graphical interface.
`mrs-md2pdf-sidecar`	Start the versioned stdio JSON-RPC engine for native desktop clients.

Standalone runtime for desktop packaging

Release engineering can build a portable sidecar runtime that includes Python, the Mardas engine, Playwright resources, and pinned Chromium. The target computer does not need Python, pip, Node.js, Git, or Chrome. Version 1.25.0 embeds this verified runtime in the first native Mardas Studio Windows installer.

BASH

```
python -m pip install -e '.[desktop]'
python -m playwright install chromium --only-shell
python scripts/build_standalone_runtime.py --clean
```

Verify the runtime manifest, protocol lifecycle, bundled browser, and a real Unicode-path PDF render:

BASH

```
python scripts/verify_standalone_runtime.py \
  build/standalone-runtime/Mardas-MD2PDF-1.25.0-runtime-windows-x86_64 \
  --render
```

The sidecar opens no localhost port. It reads one JSON-RPC request per line from `stdin`, writes protocol messages only to `stdout`, and sends logs to `stderr`. Desktop clients must call `system.health` and `system.capabilities` before rendering.

Native Mardas Studio authoring preview

Mardas Studio opens in a native Tauri window instead of a browser tab. Its Start Center provides native file selection, recent documents, and Quick Export. Version 1.25.0 also adds an authoring workspace with multiple document tabs, saved-session restore, a heading outline, literal find/replace, formatting commands, top-level front-matter controls, local assets, citations, diagnostics, and an unsaved-buffer preview.

Document saving is conflict-aware. The engine returns a revision when a file is opened, writes changes atomically, and stops instead of silently overwriting a file changed by another program. Browser storage keeps bounded recovery snapshots for dirty buffers, but recovery never saves to the source file without an explicit user action. Imported assets are restricted to supported local file types and are copied below the document's `assets/` directory.

The current editor is a dependency-free textarea foundation. It does not yet provide the complete CodeMirror workflow, and its live preview is intentionally approximate: PDF export remains authoritative for MathJax, Mermaid, pagination, fonts, and print layout. The Windows installer includes the frozen engine and pinned Chromium, so a normal user does not install Python, Playwright, Rust, or a local server.

First PDF

Create a simple PDF:

```bash
mrs-md2pdf input.md -o output.pdf
```

Create a PDF with a table of contents and the project's recommended Mardas appearance:

```bash
mrs-md2pdf input.md -o output.pdf --toc --style modern --palette emerald
--mode light
```

Create a long report with book-like page flow:

BASH

```
mrs-md2pdf input.md -o output.pdf \
  --toc \
  --toc-depth 4 \
  --toc-page-break \
  --h1-page-break \
  --style textbook \
  --palette slate \
  --mode light
```

Save the intermediate HTML for debugging:

BASH

```
mrs-md2pdf input.md -o output.pdf --debug-html output.html
```

Show the progress bar explicitly in terminal sessions:

BASH

```
mrs-md2pdf input.md -o output.pdf --progress on
```

By default, `--progress auto` shows the progress bar only when the command is running in an interactive terminal. Use `--progress off` for quiet scripts.

Recommended Workflow

A clean PDF workflow usually looks like this:

1. Write the content in Markdown.
2. Add YAML front matter for title, authors, language, direction, and cover metadata.
3. Render a first PDF with `--toc` and the desired appearance.
4. Review the cover, table of contents, math pages, code pages, image pages, and footer numbering.
5. If layout debugging is needed, export `--debug-html` and inspect the generated HTML in a browser.

6. Finalize the Markdown and render again.

For important documents, always review the final PDF visually. Automated tests can catch many problems, but typography, spacing, and page breaks still need human review.

PDF print-flow rules keep headings with following content, protect paragraphs from orphan/widow lines, and allow long code blocks or large tables to continue on the next page when keeping them together would waste space.

Front Matter

Front matter is optional YAML placed at the beginning of the Markdown file. It controls the cover page, PDF metadata, document language, document direction, and several report-specific fields.

YAML

```yaml
---
title: "My Technical Report"
subtitle: "A Markdown-powered PDF document"
authors:
  - name: "Mardas"
    email: "mragetsars@gmail.com"
    affiliation: "Mardas Lab"
    role: "Author"
summary: |
  This text appears on the cover.
  Multiline summaries are supported.
institution: "University or Organization"
department: "Department name"
course: "Course or project title"
supervisor: "Supervisor name"
date: "2026-05-20"
version: "1.25.0"
status: "Draft"
keywords: [Markdown, PDF, RTL, MathJax]
cover_label: "Technical Report"
branding:
  mode: full
brand:
  name: "Acme Research Lab"
  logo: "assets/acme-logo.svg"
  footer: "Internal Technical Report"
lang: en
dir: ltr
---
```

Common fields

Field	Purpose
`title`	Cover title and PDF metadata title.
`subtitle`	Optional text under the main title.
`author`	Simple single-author value.
`authors`	List of author objects with optional `name`, `email`, `affiliation`, and `role`.

Field	Purpose
`summary` / `description`	Cover summary and PDF metadata subject.
`institution`, `department`, `course`	Academic or organizational context.
`supervisor`, `group`, `student_id`	Optional report metadata fields.
`date`, `version`, `status`	Cover cards used for document state.
`keywords` / `tags`	Cover keywords and PDF metadata keywords.
`cover_label`	Small label above the cover title.
`branding.mode`	Cover branding mode: `off`, `subtle`, or `full`. Default is `off`.
`brand.name`, `brand.logo`, `brand.footer`	Optional organization branding shown when branding is enabled.
`cover_logo` / `logo`	Legacy/custom logo path relative to the Markdown file. The file must be a supported regular image inside the document root. Prefer `brand.logo` for new documents.
`lang`	Built-in UI language, usually `en` or `fa`.
`dir`	Document shell direction: `auto`, `ltr`, or `rtl`.

Cover behavior

The cover is rendered separately from the main document. This gives the output cleaner page numbering and better watermark behavior:

- the cover is not counted as a content page;
- footer page numbering starts after the cover;
- watermarks are applied to content pages only;
- the cover may use full-page style backgrounds;
- the cover can be disabled when a plain document is needed.

Disable the cover:

```bash
mrs-md2pdf input.md -o output.pdf --no-cover
```

The default cover is unbranded. Enable full project or organization branding only when needed:

```bash
mrs-md2pdf input.md -o output.pdf --branding full --brand-name "Acme Research Lab"
```

Use a custom brand logo:

```bash
mrs-md2pdf input.md -o output.pdf --branding full --brand-logo ./assets/logo.png
```

Keep the cover but hide the logo:

```bash
mrs-md2pdf input.md -o output.pdf --no-cover-logo
```

Use `branding.mode: subtle` for a small generated-with note, and keep `branding.mode: full` for documents that intentionally show a product or organization brand. The guides in `examples/` use full branding because they document Mardas MD2PDF itself.

Language and Direction

Direction is one of the most important parts of Persian/English PDF generation. Mardas MD2PDF separates language selection from direction control.

`lang: en` creates an English/LTR document shell, English cover labels, English callout titles, and a `Table of Contents` heading.

`lang: fa` creates a Persian/RTL document shell, Persian cover labels, Persian callout titles, and a `فهرست مطالب` heading.

Direction priority

The document direction is resolved in this order:

1. CLI `--dir rtl|ltr|auto`
2. front matter fields such as `dir`, `direction`, `text_direction`, or `document_direction`
3. language-derived default from `lang`
4. automatic detection from the Markdown body

Mixed text examples

English writing can include Persian terms such as راست به چپ, فونت فارسی, and گزارش فنی without breaking the surrounding sentence.

Persian writing can include English identifiers such as `Playwright`, `MathJax`, `GitHub Actions`, `PDF`, and `RTL/LTR` inside the same paragraph.

Inline code remains stable: `mrs-md2pdf input.md -o output.pdf --toc`.

Persian/RTL visual smoke sample

This compact sample is intentionally part of the guide because the guide is both user documentation and a live renderer test case.[1] It keeps Persian punctuation, Latin package names, Persian digits, semantic table captions, and mixed-direction cells in the official PDF examples.

1. This live sample intentionally exercises Persian punctuation, mixed Latin identifiers, Persian digits, semantic table captions, and page-local footnote rendering in the official English guide. ↩

آیا خروجی PDF برای `version 1.25.0` و شماره ۱۴۰۵ پایدار است؟ پاسخ: بله؛ جدول زیر باید RTL، mixed-script، و mixed-number hooks را فعال کند.

جدول ۱۲. نمونه جدول فارسی/RTL با عددهای ترکیبی.

بخش نمونه	مقدار	انتظار در PDF
شماره فارسی	۱۴۰۵	عدد فارسی با متن RTL پایدار بماند.
نسخه فنی	version 1.25.0 و ۱.۹.۹	Latin/Persian numerals در یک سلول خوانا بمانند.
شناسه انگلیسی	`PDF`, `TOC`, `MathJax`	identifierهای English داخل جدول فارسی جابه‌جا نشوند.

Use this section as the Persian/RTL checklist for mixed identifiers, numeric styles, RTL tables, captions, and footnote back-links. The Persian guide repeats the same concepts in an RTL document shell.

Table of Contents

Enable the table of contents:

```bash
mrs-md2pdf input.md -o output.pdf --toc
```

Control the depth:

```bash
mrs-md2pdf input.md -o output.pdf --toc --toc-depth 3
```

Start the body on a new page after the TOC:

```bash
mrs-md2pdf input.md -o output.pdf --toc --toc-page-break
```

Start each top-level heading on a new page:

```bash
mrs-md2pdf input.md -o output.pdf --h1-page-break
```

The TOC is generated from Markdown headings and preserves readable inline math in headings such as $E = mc^2$ and ϵ.

Visible TOC links and PDF viewer bookmarks use the same preserved heading destinations, so clicking either navigation layer jumps to the actual content heading rather than to a matching row inside the table of contents.

Markdown Feature Reference

Paragraphs and emphasis

Markdown paragraphs, **bold text**, *italic text*, `inline code`, links, ordered lists, unordered lists, task lists, and blockquotes are supported.

Lists

- Write content in Markdown.
- Add front matter for metadata.
- Render with the desired appearance.
- Review the final PDF.

1. Install the package.
2. Install Chromium.
3. Run the CLI.
4. Inspect the output.

Task lists

- [x] Markdown input
- [x] Persian/English typography
- [x] MathJax rendering
- [x] Code highlighting
- [ ] Final human review

Tables

Feature	Status	Notes
Mixed RTL/LTR text	Yes	Paragraphs, headings, lists, and table cells receive direction-aware styling.
Local images	Yes	Markdown and safe HTML images are embedded when possible.
MathJax	Yes	Inline and display math use separate sizing rules.
Code highlighting	Yes	Pygments is used for fenced and indented code blocks.
Mermaid diagrams	Yes	Practical `flowchart` and `graph` fences are rendered as inline SVG diagrams.
Footnotes	Yes	Multiline Markdown footnotes are supported.
Raw HTML sanitizer	Yes	Unsafe tags and event handlers are removed by default.

Blockquotes

> Good PDF publishing is more than text conversion. Typography, line height, contrast, page flow, images, formulas, and predictable rendering all matter.

Callouts

Callouts use GitHub-style markers and are translated according to the document language.

> **Note**
>
> Use callouts for explanatory notes that should stand out visually.

> **Tip**
>
> Use --debug-html output.html when you need to inspect the exact HTML sent to Chromium.

> **Warning**
>
> Use --unsafe-html only for trusted local Markdown because it disables the built-in sanitizer.

GitHub-style Markdown Compatibility

Version 1.6.2 replaces the older parallel visual controls with one appearance model: `style` controls shape and layout, `palette` controls accent colors, and `mode` controls light or dark output. Use the GitHub style when the source document is similar to a README, project guide, API note, or engineering document:

```bash
mrs-md2pdf README.md -o README.pdf --style github --palette blue --mode light --toc
```

Alerts

GitHub-style alerts are written as blockquotes and become highlighted PDF callouts:

> **Note**
>
> Notes are useful for neutral explanations.

> **Important**
>
> Important blocks should contain information the reader should not miss.

> **Caution**
>
> Caution blocks are intended for destructive or risky actions.

Autolinks

Bare URLs and email addresses are linked automatically outside code spans:

- Project page: www.example.com/mardas-md2pdf
- Maintainer contact: docs@example.com
- Literal code remains unchanged: `www.example.com`

Details and summary

HTML disclosure blocks are opened and styled for PDF output because a printed document cannot be interactive:

<details open>
<summary>Advanced conversion notes</summary>

This content is visible in the PDF and receives a printable disclosure-block style.
</details>

Heading anchors

Every heading receives a stable ID and a small permalink anchor. This improves internal links in the generated HTML and makes the PDF more navigable when links are preserved by the viewer.

Manual page breaks

For reports that need controlled pagination, use any of these forms:

```markdown
<!-- pagebreak -->
```

```markdown
:::pagebreak
:::
```

MARKDOWN

```
---page---
```

The renderer normalizes them into the same PDF page-break element.

MathJax

Inline math should match the surrounding line height: $E = mc^2$, $T = 500$, and $\Sigma = I \cdot \epsilon$ should sit naturally inside a paragraph.

Display math receives more visual space:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

A matrix example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \det(A) = -2$$

An aligned equation block:

$$\begin{aligned} \text{precision} &= \frac{TP}{TP + FP} \\ \text{recall} &= \frac{TP}{TP + FN} \end{aligned}$$

Math troubleshooting

If math appears as raw TeX:

- make sure `--no-mathjax` was not used;
- check the terminal for MathJax warnings;
- increase the browser timeout for very large math-heavy documents;
- render a smaller test file to isolate the problematic formula.

Code Blocks

Fenced code blocks support syntax highlighting and language labels.

PYTHON

```python
from dataclasses import dataclass

@dataclass
class Document:
    title: str
    lang: str = "en"


def render_message(doc: Document) -> str:
    return f"Rendering {doc.title} as PDF"

print(render_message(Document("Mardas Guide")))
```

JAVASCRIPT

```javascript
const items = ["Markdown", "Persian", "English", "MathJax", "PDF"];
const message = items.map((item, index) => `${index + 1}.
${item}`).join("\n");
console.log(message);
```

Code fences without a language are also valid:

TEXT

```
This block has no explicit language.
It should still render safely.
```

Indented code blocks are supported:

TEXT

```
SELECT title, lang, version
FROM documents
WHERE renderer = 'mardas-md2pdf';
```

Inline code is protected from math and footnote processing. For example, `$x$` and `[^note]` remain literal when they are inside backticks.

Enhanced code fences

Code fences can carry a filename/title, line numbers, and highlighted lines. This is useful when the PDF should act as a teaching or review artifact.

MARKDOWN

````
```python title="renderer.py" {2,5-6} linenos
def convert(markdown: str) -> bytes:
    html = render_markdown(markdown)
    pdf = render_pdf(html)
    metadata = inspect_pdf(pdf)
    log_export(metadata)
    return pdf
```
````

The `{2,5-6}` part means: highlight line 2 and the inclusive line range 5 through 6. Highlight numbers refer to rows inside the shown snippet, unless `linenostart` is used for source-file numbering.

The example above renders as a code block with a custom caption, line-number table, and highlighted lines:

renderer.py

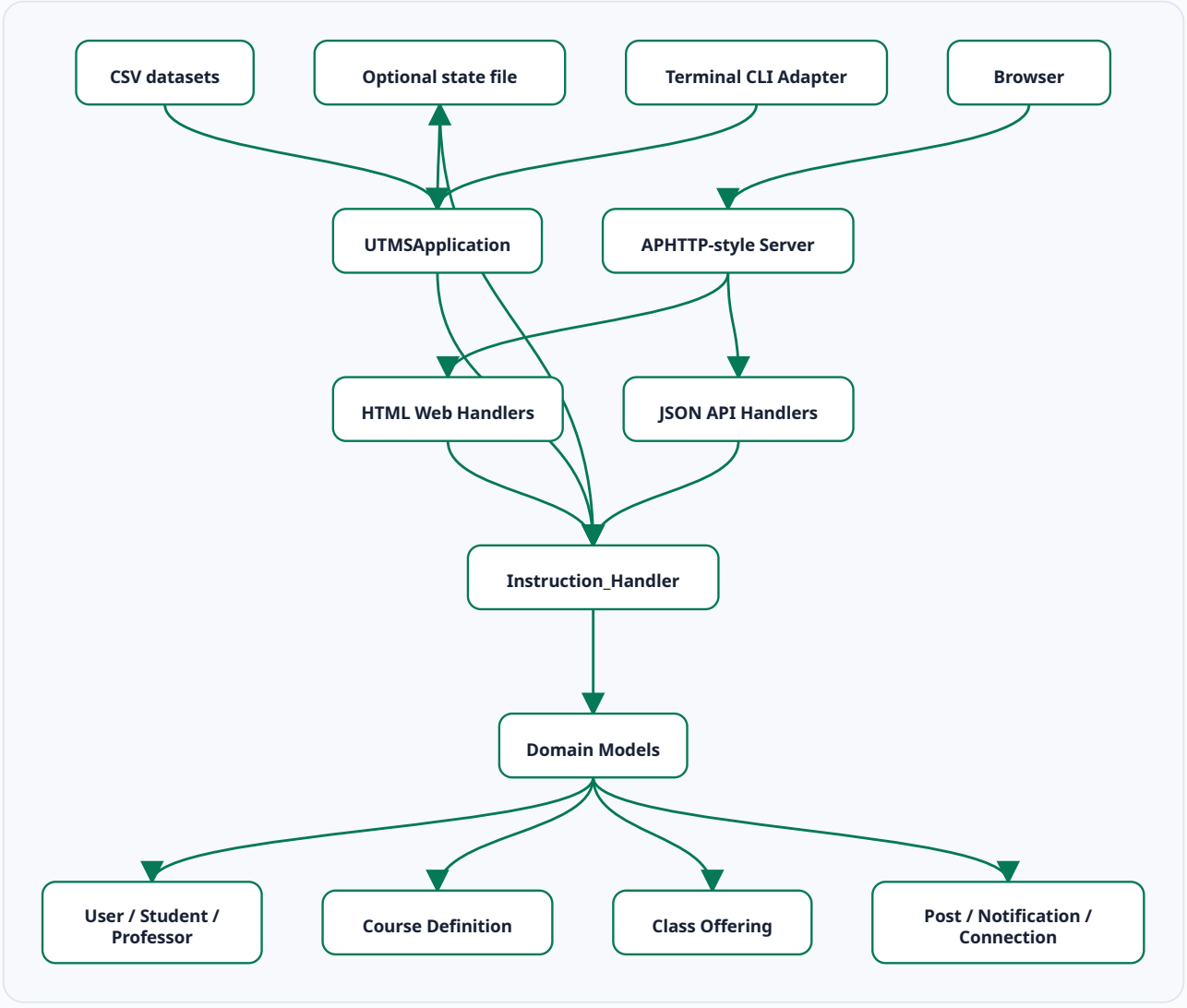
```python
def convert(markdown: str) -> bytes:
    html = render_markdown(markdown)
    pdf = render_pdf(html)
    metadata = inspect_pdf(pdf)
    log_export(metadata)
    return pdf
```

Mermaid Flowcharts

Mermaid-style flowchart fences are rendered as SVG diagrams during HTML post-processing. The renderer currently focuses on the common project-documentation subset: `flowchart` / `graph`, `TD`, `TB`, `BT`, `LR`, `RL`, rectangle nodes, rounded nodes, circles, diamonds, solid edges, dotted edges, thick edges, and labelled arrows.

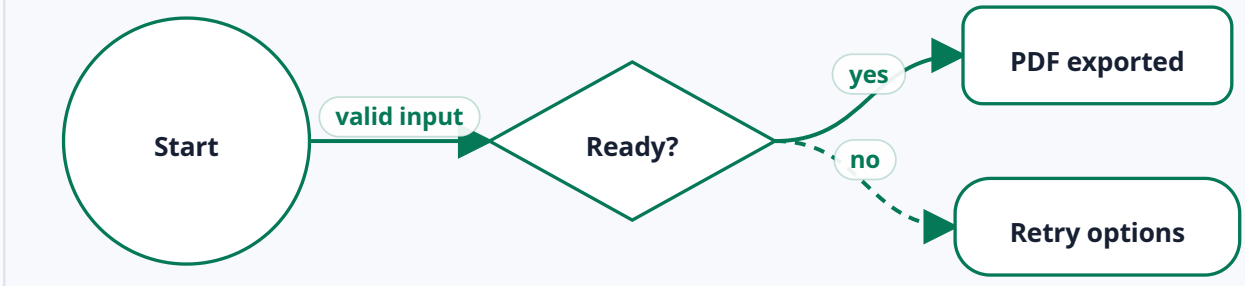
The important point for the final PDF is that the following block should appear as a diagram, not as source code:

MERMAID



A smaller labelled-edge sample is useful when checking edge labels and node shapes:

MERMAID



If a Mermaid block uses advanced syntax outside the supported subset, keep a small fallback screenshot or image in the Markdown until that syntax is added to the renderer. For ordinary architecture and data-flow diagrams, the built-in renderer is enough and does not need internet access.

Images and Safe HTML

Markdown images are embedded when they point to local files. A common caption pattern is also promoted to a real PDF figure:

```markdown
![Architecture diagram](images/architecture.png)

*Figure 1. Architecture overview.*
```

When the caption starts with `Figure`, `Fig.`, `شكل`, or `تصویر`, the renderer wraps the image and caption in a semantic figure block. Safe HTML image attributes such as `width` and `height` are preserved:

```html
<img src="images/architecture.png" width="760" alt="Architecture diagram with explicit width">
```

Images are resolved relative to the Markdown file. Local images are embedded into the generated HTML/PDF when they are small enough to embed safely.

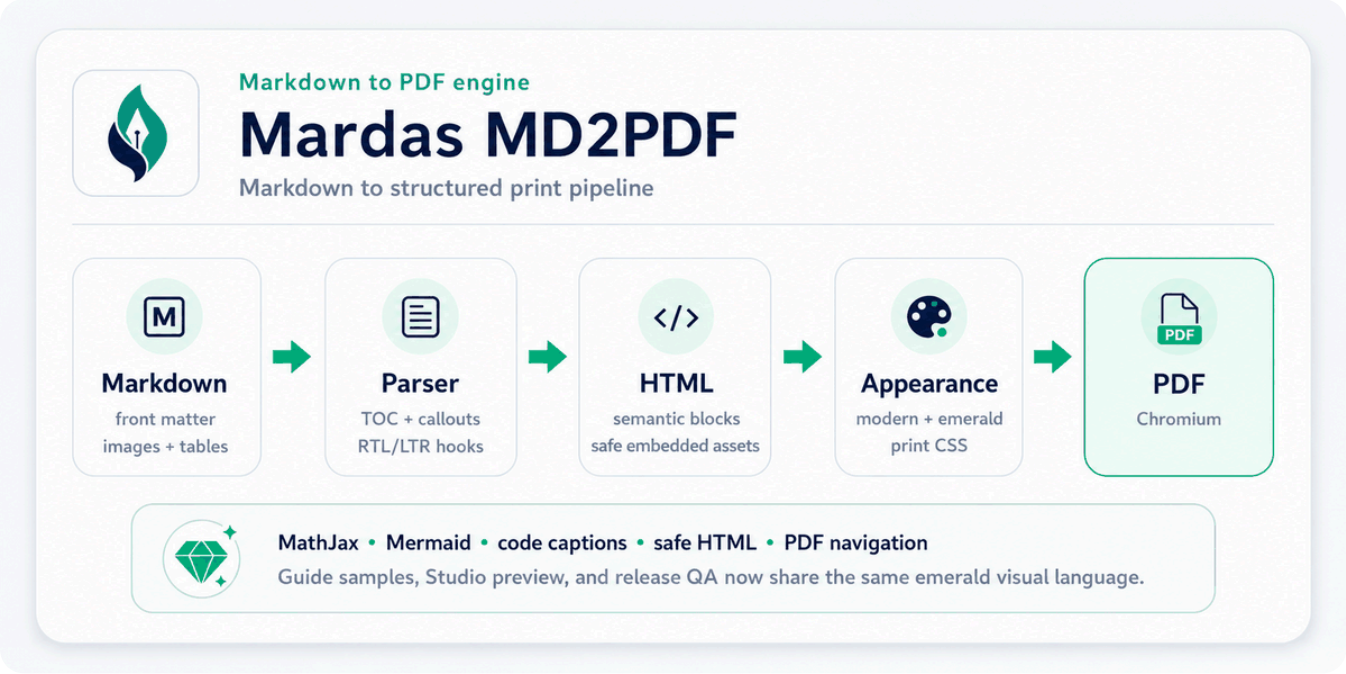


Figure 1. Architecture overview.

Safe raw HTML images can also be used when explicit sizing is needed. The guide deliberately reuses the same banner/architecture asset here so the HTML example exercises width handling without switching to an unrelated raw logo:

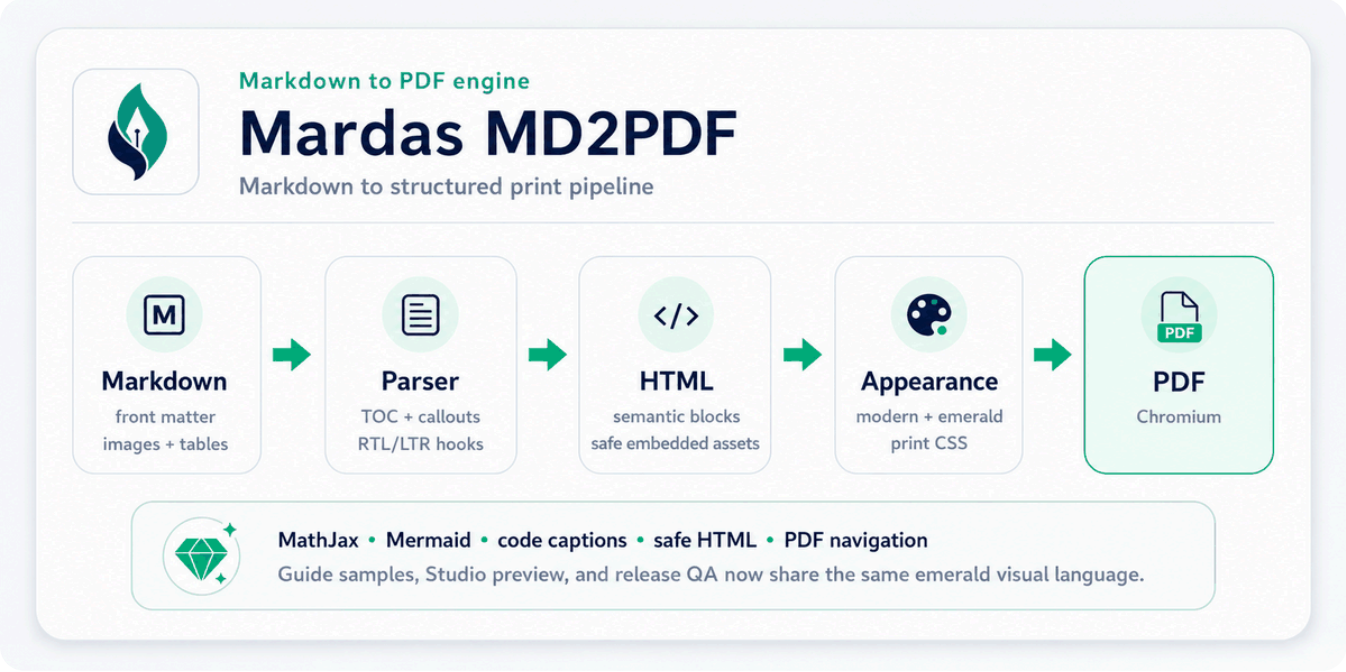


Image rules

- Prefer local images for stable PDF output.
- Keep images reasonably sized before embedding them.

- Use paths relative to the Markdown file.
- Keep image paths inside the Markdown document directory. Absolute paths, `file:` URLs, parent-directory escapes such as `../secret.png`, and current-working-directory fallbacks are blocked by default.
- If an image cannot be embedded safely, it is replaced with a visible blocked placeholder so Chromium does not load it through the document `<base>` URL.
- Remote `http(s)` images are blocked by default for privacy and reproducibility. Use `--allow-remote-assets` only for trusted documents that intentionally fetch network images.
- In the GUI, attach local image files or image folders before exporting.
- Very large images are blocked and a warning is printed to avoid excessive memory usage.

Safe HTML

Raw HTML is sanitized by default. The sanitizer keeps document-oriented elements such as `<div>`, `<span>`, `<table>`, `<figure>`, and `<img>`, while removing active or unsafe content such as scripts, event handlers, iframes, forms, remote stylesheets, `file:` URLs, unsafe URL schemes, and non-raster `data:` image URLs.

Safe `data:` images are limited to common raster formats such as PNG, JPEG, GIF, WebP, BMP, and AVIF. Inline SVG data URLs are blocked in safe HTML; use a local SVG/PNG file you trust or a Mermaid block for diagrams.

Use unsafe HTML only for trusted local files:

BASH

```
mrs-md2pdf input.md -o output.pdf --unsafe-html
```

Security and Trusted Inputs

Mardas MD2PDF is a local publishing tool. Treat Markdown, GUI attachments, cover logos, watermarks, and raw HTML as trusted author content unless you run the converter inside an isolated environment.

Default rendering keeps a conservative file boundary: local images are resolved relative to the Markdown file, embedded as `data:` URLs when safe, and blocked with a visible placeholder when they point outside the document directory or cannot be embedded. Remote `http(s)` images are also blocked by default and require `--allow-remote-assets`. Raw HTML is sanitized unless `--unsafe-html` is used.

Chromium sandboxing is controlled by `--chromium-sandbox`:

Mode	Behavior
`auto`	Keep sandboxing on for normal users; disable only when running as root.
`on`	Always request Chromium sandboxing.
`off`	Pass `--no-sandbox`; use only in trusted containers or isolated CI jobs.

For untrusted documents, render in a container or disposable environment and keep `--unsafe-html` off. See `../SECURITY.md` for the full policy.

PDF navigation and metadata

Rendered PDFs include standard document metadata from front matter and a viewer outline generated from Markdown headings. The printed table of contents remains visible in the document body, while the PDF outline gives readers a sidebar-friendly way to jump between sections in compatible PDF viewers.

Use `--toc` when you also want a printed table of contents. The PDF outline is generated from the same heading collection so both navigation surfaces stay in sync.

Input and output integrity

Malformed, recursive, excessively deep, or oversized YAML front matter is rejected with a controlled diagnostic. UTF-8 files with a BOM are accepted. The CLI also refuses to use the same underlying file for Markdown input, PDF output, or debug HTML, and commits successful PDF/debug output atomically so an interrupted write preserves the previous artifact.

Relative links to local filesystem targets remain readable in the document but are not exported as machine-specific `file:` links. Manual and generated heading identifiers are deduplicated so TOC and PDF destinations remain unambiguous.

Page Flow and Layout

Manual page breaks

A manual page break can be inserted with safe HTML:

```HTML
<div class="md2pdf-page-break"></div>
```

Page size

Use a named page size:

```BASH
mrs-md2pdf input.md -o output.pdf --page-size A4
```

Use landscape orientation:

```BASH
mrs-md2pdf input.md -o output.pdf --page-size "A4 landscape"
```

Use explicit dimensions:

```bash
mrs-md2pdf input.md -o output.pdf --page-size "210mm 297mm"
```

Named A0–A6, B0–B6, Letter, Legal, Tabloid, and Ledger sizes are converted to explicit Chromium dimensions so unsupported named formats do not silently fall back to A4. Custom dimensions must remain between 10 mm and 5000 mm per side. Unknown or out-of-range values fail early, and Studio uses the same validation with a structured `invalid_page_size` error.

Margins

```bash
mrs-md2pdf input.md -o output.pdf \
  --margin-top 18mm \
  --margin-bottom 18mm \
  --margin-x 16mm
```

Watermarks

Text watermark:

```bash
mrs-md2pdf input.md -o output.pdf --watermark "DRAFT"
```

Image watermark:

```bash
mrs-md2pdf input.md -o output.pdf \
  --watermark-image ./src/mardas_md2pdf/assets/mardas-md2pdf-logo.png \
  --watermark-opacity 0.05 \
  --watermark-width 95mm
```

Watermarks are applied to content pages only, not to the cover page.

Cover Branding

Generated PDFs are unbranded by default. This avoids turning ordinary reports into product advertisements and keeps the cover focused on the document owner.

```yaml
branding:
  mode: off
```

Available modes:

Mode	Use case
`off`	Default for ordinary reports and handouts.
`subtle`	Small generated-with note for informal shared drafts.
`full`	Intentional project or organization branding.

Custom organization branding:

YAML

```yaml
branding:
  mode: full
brand:
  name: "Acme Research Lab"
  logo: "assets/acme.png"
  footer: "Internal Technical Report"
```

CLI equivalent:

BASH

```bash
mrs-md2pdf input.md -o output.pdf \
  --branding full \
  --brand-name "Acme Research Lab" \
  --brand-logo assets/acme.png \
  --brand-footer "Internal Technical Report"
```

The English and Persian guides explicitly use `branding.mode: full` because they are project examples.

Appearance

Mardas MD2PDF uses one appearance system instead of parallel visual presets. Choose a `style` for document shape, a `palette` for accent colors, and a `mode` for light or dark output.

Style	Best for
`modern`	General documentation, proposals, software reports, and product-style guides.
`github`	README-like project documentation and GitHub-style Markdown output.
`textbook`	Long educational documents, course notes, Persian/English learning material.
`academic`	Formal reports, university documents, thesis-like drafts, and structured papers.

Palette	Accent feel
`blue`	Default professional blue.
`emerald`	Calm green.
`violet`	Creative purple.
`amber`	Warm teaching/review accent.
`rose`	Editorial red-pink accent.
`slate`	Cool understated neutral.
`neutral`	Minimal grayscale.

Choose appearance from the CLI:

```bash
mrs-md2pdf input.md -o output.pdf --style modern --palette emerald --mode light
mrs-md2pdf input.md -o output.pdf --style academic --palette emerald --mode dark
```

Or store it in front matter:

```yaml
appearance:
  style: modern
  palette: emerald
  mode: light
```

Discover choices with `--list-styles`, `--list-palettes`, and `--list-modes`.

Dark mode is style-aware. `modern` uses a deep navy surface, `github` follows a GitHub-like dark surface, `textbook` keeps the nearly black screen/cover look, and `academic` uses a warm charcoal surface. Palettes still control accent colors in both light and dark modes, including cover decorations and callouts.

Publication Quality Profiles

Normal interactive work uses the `standard` profile. It preserves the historical behavior: quality problems are recorded and emitted as warnings so exploratory exports can continue. For a thesis, journal submission, public report, or release artifact, use `strict-publication` so the build fails when the renderer cannot prove that formulas, required fonts, and internal PDF navigation survived the complete pipeline.

BASH

```
mrs-md2pdf report.md -o report.pdf \
  --quality-profile strict-publication \
  --require-font Vazirmatn \
  --require-font "JetBrains Mono" \
  --quality-report build/report-quality.json
```

The report is bounded JSON suitable for CI. It records the selected profile, pass/warning/error counts, MathJax expressions detected and rendered, unresolved formula nodes, required-font availability, loaded browser font faces, hashes of trusted font files supplied through `--font-dir`, named destinations copied or reconstructed, internal link rewrites, and the final artifact status. It does not embed font files or document contents.

Each category can be overridden independently:

BASH

```
mrs-md2pdf report.md -o report.pdf \
  --quality-profile strict-publication \
  --math-error-policy error \
  --font-error-policy warn \
  --navigation-error-policy error
```

`error` stops the build, `warn` records the problem and continues, and `ignore` records that the condition was deliberately ignored. When a category option is omitted, it inherits the profile: `warn` for `standard` and `error` for `strict-publication`. Strict publication defaults to requiring `Vazirmatn`

when no family is specified. Mardas MD2PDF does not redistribute font binaries; install trusted fonts on the machine or use `--font-dir`.

The same settings are available in `mardas.toml`:

```toml
[quality]
profile = "strict-publication"
# Omit category policies to inherit strict `error` behavior.
required_fonts = ["Vazirmatn", "JetBrains Mono"]
report = "build/report-quality.json"
```

Studio exposes these controls under **Publication Quality**. A queued export returns a bounded quality summary after completion, and strict failures use the same stable backend error path as other render failures. PDF-like preview remains non-authoritative; the final export and its quality report are the release evidence.

Project Configuration and Diagnostics

Use a versioned `mardas.toml` when a document or repository needs repeatable settings without a long command line. Create the initial file in the current directory:

```bash
mrs-md2pdf init
```

The generated configuration is deliberately conservative: safe HTML, blocked remote assets, an ordinary A4 page, explicit appearance settings, and schema version `1`. Relative paths in the configuration are resolved from the directory containing `mardas.toml`, not from the shell working directory.

A compact project configuration can look like this:

TOML

```toml
schema_version = 1

[project]
title = "Numerical Methods Report"
author = "Research Team"
direction = "auto"

[output]
page_size = "A4"
toc = true
toc_depth = 3
toc_page_break = true
cover = true
header_footer = true
mathjax = true
margin_top = "18mm"
margin_bottom = "20mm"
margin_x = "16mm"

[appearance]
style = "academic"
palette = "emerald"
mode = "light"

[branding]
mode = "off"
# logo = "assets/lab-logo.png"

[security]
unsafe_html = false
allow_remote_assets = false

[browser]
chromium_sandbox = "auto"
timeout_ms = 120000

[quality]
profile = "standard"
# math_errors = "warn"
# font_errors = "warn"
# navigation_errors = "warn"
# required_fonts = ["Vazirmatn"]
# report = "build/render-quality.json"
```

The CLI discovers the nearest `mardas.toml` by starting beside the Markdown file and walking toward the filesystem root. Use `--config path/to/mardas.toml` for an explicit file or `--no-config` to disable discovery. Precedence is deterministic:

```text
explicit CLI option > mardas.toml > equivalent front matter > built-in default
```

Negative overrides such as `--no-toc`, `--cover`, `--header-footer`, `--mathjax`, `--safe-html`, and `--block-remote-assets` make it possible to override enabled or disabled project settings from automation scripts.

Validate configuration and Markdown without launching Chromium:

```bash
mrs-md2pdf validate report.md
mrs-md2pdf validate report.md --format json
```

Validation reports malformed TOML or YAML, unsupported schema versions, unknown keys, invalid values, missing configured assets, blocked images, heading-level jumps, and risky security settings. Diagnostics use stable identifiers such as `MARDAS-E103`, `MARDAS-E109`, `MARDAS-W203`, and `MARDAS-W301`; automation should rely on the code rather than matching the English message.

Inspect the effective values and their sources:

```bash
mrs-md2pdf explain-config report.md
mrs-md2pdf explain-config report.md --format json
```

Check the local runtime, packaged MathJax, required dependencies, Chromium discovery, configuration, and optionally the document itself:

BASH

```
mrs-md2pdf doctor
mrs-md2pdf doctor report.md --format json
```

Warnings do not make `validate` or `doctor` fail, while error diagnostics return a non-zero exit status. A configuration that enables `unsafe_html` or remote assets is accepted only as an explicit project decision and is surfaced as a warning because it expands the trust boundary and can reduce build reproducibility.

Multi-file Book Mode

Book Mode builds one PDF from an explicit, deterministic list of Markdown chapters. It is intended for books, theses, long reports, course notes, and documentation sets whose source should remain split into manageable files.

Create a starter project:

BASH

```
mrs-md2pdf init my-book --book
```

The generated project contains `mardas.toml` and two starter chapter files:

TEXT

```
my-book/
├── mardas.toml
└── chapters/
    ├── 01-introduction.md
    └── 02-content.md
```

A Book Mode manifest uses an ordered `[book].chapters` array. The array order, not directory sorting, controls the PDF order:

TOML

```toml
schema_version = 1

[project]
title = "Numerical Methods Handbook"
author = "Research Group"
direction = "ltr"

[output]
toc = true
toc_depth = 4
cover = true
header_footer = true
mathjax = true

[book]
chapters = [
  "chapters/01-introduction.md",
  { path = "chapters/02-methods.md", title = "Methods and Experiments" },
  "chapters/03-results.md",
]
output = "dist/handbook.pdf"
chapter_page_break = true
```

The optional chapter `title` replaces the visible level-one heading and global TOC title for that chapter without modifying the Markdown source. If a chapter has no level-one heading, Book Mode generates one from the chapter title and reports `MARDAS-W501`.

Validate and inspect the complete project without launching Chromium:

BASH

```bash
mrs-md2pdf validate-book my-book
mrs-md2pdf validate-book my-book --format json
mrs-md2pdf explain-book my-book --format json
```

Build the PDF and optionally keep the combined renderer HTML:

```bash
mrs-md2pdf build-book my-book
mrs-md2pdf build-book my-book -o releases/handbook.pdf
mrs-md2pdf build-book my-book --debug-html build/handbook.html --progress on
```

Book Mode applies these rules:

1. Every chapter path must be relative to `mardas.toml`, remain inside the project root after symlink resolution, use a supported Markdown extension, and appear only once.
2. Project settings provide the shared cover, output, appearance, security, browser, font, watermark, and PDF configuration. Chapter front matter still controls chapter-local language and content metadata where applicable.
3. Images may be stored beside a chapter or in a shared directory inside the project root, such as `assets/`. Files outside the project root remain blocked.
4. Heading, anchor, and footnote IDs are namespaced per chapter before assembly, so repeated headings such as `# Introduction` remain unambiguous.
5. Relative Markdown links to another listed chapter are converted to internal PDF links. A fragment such as `02-methods.md#model` resolves to the namespaced heading in that chapter. Other local file links remain inert for portable PDF output.
6. A page break is inserted between chapters when `chapter_page_break = true`.
7. The final PDF contains one cover, one global TOC, one nested PDF outline, one metadata set, stable named destinations, and one output artifact.

Book Mode shares the same reference symbol table across all listed chapters. With `numbering_scope = "chapter"`, an object in chapter 2 receives a number such as `2.1`; with `global`, numbering remains continuous across the complete book. Bibliography processing remains a separate phase.

Cross-references and Numbering

Cross-references are opt-in. Enable them in `mardas.toml` or front matter:

```toml
[references]
enabled = true
numbering_scope = "global"
list_of_figures = true
list_of_tables = true
list_of_equations = true
list_of_listings = true
```

The same fields may be written under a front-matter `references:` mapping. Command-line automation can override them with `--references`, `--no-references`, `--numbering-scope global|chapter`, and paired `--list-of-*` / `--no-list-of-*` options.

Supported semantic object kinds are:

Table 1 Supported semantic reference objects

Prefix	Object	Label placement
`fig`	Markdown images and Mermaid figures	In the caption or as a standalone marker immediately after the object.
`tbl`	Markdown tables	At the end of the `Table:` caption.
`eq`	Display MathJax equations	As a standalone marker immediately after the equation.
`lst`	Fenced code listings	In the opening fence metadata.

Use a semantic label and refer to it with the matching `@kind:name` token:

MARKDOWN

```
See @fig:processing-flow, @tbl:metrics, @eq:energy, and @lst:training-
loop.

![Architecture](assets/architecture.svg)

*Figure. Processing architecture.* {#fig:processing-flow}

| Metric | Value |
| :--- | ---: |
| Accuracy | 0.98 |

Table: Evaluation metrics {#tbl:metrics}

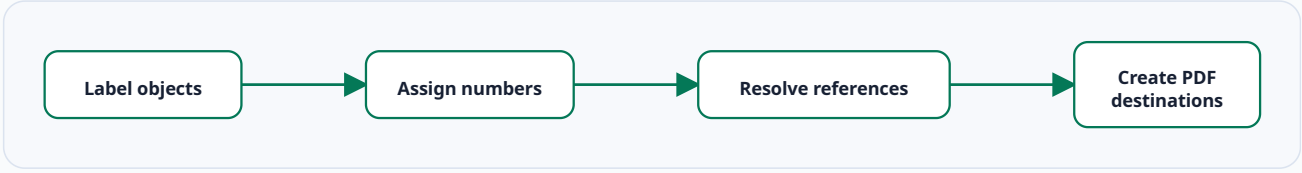
$$
E = mc^2
$$

{#eq:energy}

```python title="Training loop" {#lst:training-loop}
print("train")
```
```

The next examples are live renderer checks in this guide. The references **Figure 1**, **Table 1**, **Equation 1**, and **Listing 1** must become internal PDF links with stable localized numbers.

FIGURE 1 REFERENCE PROCESSING FLOW



$$E = mc^2 \tag{1}$$

Listing 1 Reference validation

```
labels = {"fig:flow", "tbl:metrics"}
assert len(labels) == 2
```

Labels must be unique across the complete output. Book Mode resolves references only after all chapters are assembled, so a reference in one chapter can target a labeled object in another. Duplicate labels, unresolved references, kind mismatches, malformed labels, and unattached label markers produce stable `MARDAS-E60x` / `MARDAS-W603` diagnostics before Chromium starts. References inside code, existing links, scripts, styles, and other literal contexts remain untouched.

Generated reference lists are placed after the table of contents and before the document body. Only explicitly labeled objects are included. This guide enables all four lists as a live release sample.

Bibliography and Citations

The citation engine is offline-first and opt-in. It reads local BibTeX (`.bib`) or CSL JSON (`.json`) files and never performs DOI lookup or network metadata retrieval during a build.

Enable it in `mardas.toml`:

TOML

```toml
[bibliography]
enabled = true
sources = ["references.bib"]
style = "author-date"
include_uncited = false
# title = "References"
```

The same source may be declared in front matter:

YAML

```yaml
bibliography: references.bib
citations:
  enabled: true
  style: author-date
  include_uncited: false
```

Supported citation forms are deliberately small and deterministic:

MARKDOWN

```markdown
Prior work supports this result [@doe2024].
Several studies agree [@doe2024, p. 12; @smith2023].
Narrative citation: @doe2024 describes the method.
```

Use `style = "numeric"` for first-use numbering such as `[1]`; `author-date` generates localized author/year citations and deterministic `a`, `b`, ... suffixes when the same author has multiple works in one year. Bibliography entries are linked to their first citation and citation links target stable `#bib-*` destinations. Book Mode loads project bibliography sources once and resolves all chapter citations only after chapter assembly, so one bibliography is generated for the complete book.

The next sentence is a live renderer check. It must produce two disambiguated grouped citations (**Rastegar, 2026a, p. 14**; **Rastegar, 2026b**) and one narrative citation to **احمدی (1404)**, followed by one generated bibliography at the end of this guide.

Validation rejects missing sources, invalid BibTeX/CSL JSON, duplicate keys, undefined citation keys, malformed citation groups, repeated source files, oversized sources, and excessive entry counts with stable `MARDAS-E70x` diagnostics before Chromium starts. Citation-like text inside code, existing links, scripts, styles, and other literal contexts remains unchanged.

Performance and Large Documents

Repeated Studio exports use a bounded queue and a persistent renderer session. Each worker may reuse one Chromium process, but every export receives a fresh browser context so page state, cookies, object URLs, and storage are isolated. The Studio footer displays the real render stage and percentage instead of an estimated timer. Use **Cancel** to cancel a queued job immediately or request cooperative cancellation of a running job.

Tune the local queue from the GUI command line:

```bash
mrs-md2pdf-gui \
  --render-workers 2 \
  --export-queue-size 6 \
  --render-idle-timeout 60
```

`--render-workers` controls simultaneous Chromium workers, `--export-queue-size` bounds waiting jobs, and `--render-idle-timeout` closes an unused persistent browser. More workers can increase throughput but also increase peak memory; keep the default unless measurements justify a change.

For reproducible before/after measurements, run:

```bash
python scripts/benchmark_large_documents.py \
  --profiles small,pages50,pages250,pages500,editor-loop \
  --mode both \
  --repeats 3 \
  --output-dir build/performance
```

The `cold` mode launches Chromium for each conversion. The `session` mode keeps Chromium alive and creates a new browser context per repeat. Compare the JSON report's wall-clock distribution, page count, PDF size, browser-launch count, and peak RSS in the same environment. A single fast run is not sufficient evidence of a performance improvement.

Completed Studio PDFs remain in private bounded temporary storage only until download or expiry and are streamed to the browser. Cancellation is cooperative: a currently executing Chromium `page.pdf()` call completes before the next safe cancellation checkpoint.

Accessibility and Archival Readiness

Declare the document language explicitly so HTML, PDF metadata, assistive tools, and localized labels receive a stable language signal:

```toml
[project]
language = "en-US"
direction = "ltr"
```

The same setting can be supplied in front matter with `lang: en-US` or for one conversion with `--lang en-US`.

Run the source audit before opening Chromium:

```bash
mrs-md2pdf audit-accessibility report.md
mrs-md2pdf audit-accessibility report.md --format json --fail-on warning
mrs-md2pdf audit-book-accessibility path/to/book
```

The source audit checks heading hierarchy, alternative text, link names, table headers and captions, declared language, and the selected appearance contrast. It reports stable `MARDAS-A...` diagnostic codes with file and line information where available.

Use descriptive text and semantic table structure:

MARKDOWN

```
![Pipeline from Markdown input to PDF output](images/architecture.png)

*Figure. Publishing pipeline overview.*

| Stage | Accessible output |
| :--- | :--- |
| Source audit | Structured diagnostics |
| PDF audit | Metadata and font readiness report |

Table: Accessibility verification stages

Read the [security and accessibility guidance](../SECURITY.md).
```

Rendered image figures are associated with their captions. Table captions receive deterministic identifiers, column/row headers receive `scope`, and tables reference their captions. These HTML improvements support browser semantics, but they do not by themselves create a verified tagged PDF.

Inspect the final artifact separately:

BASH

```
mrs-md2pdf audit-pdf output.pdf --profile accessibility
mrs-md2pdf audit-pdf output.pdf --profile archival
mrs-md2pdf audit-pdf output.pdf --profile all --format json --fail-on
never
```

Generated PDFs include catalog `/Lang`, viewer title preferences, document information metadata, and XMP metadata. The PDF audit reports font embedding, ToUnicode maps, tagging signals, output intents, JavaScript, attachments, and PDF/A identifiers. Current Chromium output is not claimed to be PDF/UA compliant, and the project does not add a false structure tree or PDF/A identifier. Formal conformance requires an independent validator and manual accessibility review.

GUI Workflow

Launch the GUI for a standalone document:

```bash
mrs-md2pdf-gui
```

Open a live project workspace rooted at `mardas.toml`:

```bash
mrs-md2pdf-gui --project path/to/project
```

Project Workspace mode shows the ordered Book Mode chapters and supported project text files in a Project Explorer. The Problems panel uses the same `MARDAS-*` diagnostics as `validate` and `validate-book`; selecting a problem opens the responsible file and moves the editor to its line and column when available. The active Markdown file uses renderer-backed project configuration, bibliography, cross-references, and project-root asset rules. **Preview Book** and **Export Book** operate on the saved complete project, so save the active file before running either action.

Project files are edited in place only after an explicit `--project` launch. Studio accepts supported UTF-8 text files inside the project root, rejects hidden/generated paths and symbolic links, limits each editable file to 4 MiB, and uses the previously opened SHA-256 value to reject an external-change conflict. Successful saves use atomic replacement and preserve the existing file mode. The legacy **Open Bundle** and **Save Bundle** controls remain available for portable `.mardas.json` snapshots; a bundle is not the same as a live project workspace.

The GUI is useful for users who prefer a visual workflow:

1. Paste or type Markdown.
2. Fill the **Document** section for title, author, output filename, page size, and direction.
3. Choose **Appearance** cards for style, palette, and light/dark mode.
4. Choose **Branding** only when the PDF should show a product, organization, or lab mark.
5. Use **Layout** for TOC, cover, and page-flow choices.
6. Open **Advanced** only when you need watermarks, hidden page numbers, or attached local assets.

7. Use the default PDF-like preview for renderer-backed HTML with page size, margins, and auto-fit scaling; switch to Fast preview only when you need instant browser-local feedback and can accept approximate Markdown parsing. Fast Preview does not fetch remote/local image paths and disables unsafe or filesystem link schemes.
8. Use **Ctrl/Cmd+S** to save Markdown or the active Project Workspace file, **Ctrl/Cmd+Shift+S** to save a portable `.mardas.json` bundle, and **Ctrl/Cmd+Enter** to export quickly.

Studio stores the current draft, layout, interface mode, direction toggle, editor width, and export settings in browser local storage. This makes accidental refreshes less disruptive during long editing sessions. Use **Reset State** when you want to clear the saved local draft and return to a clean workspace. This section is the Studio workflow reference for users.

If an export fails, Studio shows the HTTP status and stable backend error code, such as `invalid_json`, `invalid_page_size`, `invalid_toc_depth`, `invalid_watermark_opacity`, `markdown_too_large`, or `render_failed`. If you bind Studio to a non-local host, the backend prints a warning because other users on the reachable network can submit Markdown and attached assets.

> **Important**
>
> Studio defaults to an auto-scaling PDF-like renderer preview with page size and margins. Fast Preview remains useful for instant editing, but it is an approximate browser-local parser with a deliberately restricted URL policy; final fidelity, page breaks, attached assets, MathJax, Mermaid, and security validation are determined by the backend renderer and Chromium print layout during PDF export.

CLI Reference

Option	Description
`input`	Input Markdown file.
`-o`, `--output`	Output PDF path.
`--title`, `--author`, `--description`	Override metadata from front matter.
`--toc`, `--toc-depth`	Enable and configure the table of contents.
`--references`, `--no-references`	Enable or disable semantic object numbering and cross-references.
`--numbering-scope`	Use continuous `global` numbering or `chapter` numbering in Book Mode.

Option	Description
`--list-of-figures`, `--list-of-tables`, `--list-of-equations`, `--list-of-listings`	Generate selected reference lists; each option also has a paired `--no-list-of-*` override.
`--citations`, `--no-citations`	Enable or disable citation resolution and generated bibliography output.
`--bibliography PATH`	Add a local `.bib` or CSL `.json` source; repeat for multiple sources.
`--citation-style`	Select the built-in `author-date` or `numeric` style.
`--bibliography-title`, `--include-uncited`, `--cited-only`	Customize the bibliography heading and whether uncited entries are included.
`--toc-page-break`, `--h1-page-break`	Control printed page flow.
`--style`	Choose `modern`, `github`, `textbook`, or `academic`.
`--palette`	Choose `blue`, `emerald`, `violet`, `amber`, `rose`, `slate`, or `neutral`.
`--mode`	Choose `light` or `dark`.
`--list-styles`, `--list-palettes`, `--list-modes`	Print available appearance choices and exit.
`--page-size`	Use `A4`, `Letter`, `Legal`, `A4 landscape`, or dimensions such as `210mm 297mm`.
`--dir`	Force `auto`, `ltr`, or `rtl`.
`--margin-top`, `--margin-bottom`, `--margin-x`	Control page margins.
`--font-dir`	Directory containing local Vazirmatn font files.
`--chromium-path`	Custom Chromium or Chrome executable path.
`--chromium-sandbox`	Chromium sandbox mode: `auto`, `on`, or `off`. Default: `auto`.
`--debug-html`	Save the intermediate HTML.
`--no-cover`, `--branding`, `--brand-name`, `--brand-logo`, `--brand-footer`, `--no-cover-logo`	Configure the cover and explicit branding.
`--watermark`, `--watermark-image`	Add mode-aware text or image watermarks over content pages.
`--allow-remote-assets`	Allow trusted Markdown to fetch remote `http(s)` images. Disabled by default.
`--no-header-footer`	Disable printed footer.
`--no-mathjax`	Do not load MathJax.
`--unsafe-html`	Disable raw HTML sanitization for trusted files.
`--timeout-ms`	Browser timeout in milliseconds.
`--progress`	Terminal progress bar mode: `auto`, `on`, or `off`. Default: `auto`.

Run the full help command when needed:

BASH

```
mrs-md2pdf --help
```

Automation and CI

A typical automation command can be as simple as:

BASH

```
mrs-md2pdf docs/report.md -o build/report.pdf --toc --style modern --
palette emerald --mode light
```

The repository includes a GitHub Actions CI workflow that runs Ruff, pytest, and a Chromium render smoke test on supported Python versions. For CI workflows, prefer explicit options:

BASH

```
mrs-md2pdf docs/report.md -o build/report.pdf \
  --toc \
  --toc-depth 4 \
  --style modern \
  --palette emerald \
  --mode light \
  --page-size A4 \
  --dir auto \
  --timeout-ms 60000 \
  --progress off
```

For debugging CI failures, save the intermediate HTML as an artifact:

BASH

```
mrs-md2pdf docs/report.md -o build/report.pdf --debug-html
build/report.html
```

Verifying Release Artifacts

Official release artifacts include a deterministic wheel, source distribution, English/Persian guide PDFs, an SPDX 2.3 runtime SBOM, `RELEASE-MANIFEST.json`, and `CHECKSUMS.sha256`. Platform-specific offline Python bundles are built separately on Linux, Windows, and macOS.

Verify checksums before installation:

```bash
sha256sum -c CHECKSUMS.sha256
```

Verify the release inventory and SBOM structure from a repository checkout:

```bash
python scripts/finalize_release_artifacts.py \
  --artifact-dir path/to/release \
  --version X.Y.Z \
  --require-sbom \
  --verify-only
```

A GitHub-hosted artifact can also be checked against signed build provenance:

```bash
gh attestation verify artifact-name \
  --repo mragetsars/Mardas-MD2PDF
```

After extracting an offline Python bundle, run:

```bash
python install.py --target mardas-md2pdf-venv
```

The bundle installs Python wheels without contacting a package index. It does not include Chromium or a standalone Python runtime; browser installation remains a separate, explicit step.

Troubleshooting

Chromium is missing

Run:

```bash
python -m playwright install chromium
```

or pass a custom browser path:

```bash
mrs-md2pdf input.md -o output.pdf --chromium-path /path/to/chrome
```

Persian text looks wrong

Install a Persian-capable font such as Vazirmatn, or pass a font directory:

```bash
mrs-md2pdf input.md -o output.pdf --font-dir ./fonts
```

If the directory is missing or does not contain recognized font files, the renderer prints a fallback warning and uses system fonts.

Images do not appear

Check that image paths are relative to the Markdown file and stay inside the Markdown document directory. Absolute paths, `file:` URLs, parent-directory escapes, missing files, remote images, and very large files are replaced with a visible blocked placeholder instead of being loaded by Chromium. If you use the GUI, attach the local images or folders before export and check renderer warnings.

Math appears as raw TeX

Make sure MathJax is enabled and check for renderer warnings. For large documents, increase the timeout:

```bash
mrs-md2pdf input.md -o output.pdf --timeout-ms 90000
```

Layout needs inspection

Export debug HTML:

```bash
mrs-md2pdf input.md -o output.pdf --debug-html output.html
```

Open `output.html` in a browser and inspect the generated structure and CSS.

PDF Preflight Checks

For important public PDFs, run a quick preflight after building the examples. This check rasterizes representative pages, lists embedded fonts, and records PDF syntax warnings before a document is released:

```bash
python scripts/check_pdf_preflight.py \
  examples/GUIDE.en.pdf \
  examples/GUIDE.fa.pdf \
  --pages 1,2,3 \
  --output build/pdf-preflight.json
```

A clean visual result still matters more than a raw tool warning, but preflight output gives maintainers a repeatable place to catch font, rasterization, and PDF parser regressions.

Footnotes

Footnotes are useful for references, technical notes, and extra explanations.[2]

2. Mardas MD2PDF intentionally uses Chromium for layout instead of drawing every paragraph directly on a PDF canvas. This gives the project strong support for CSS print rules, mixed direction text, MathJax SVG output, tables, local images, and syntax-highlighted code. ↩

   - Page-local footnote blocks are inserted near the reference instead of being collected as document-end endnotes.
   - Multiline footnotes are supported.
   - Markdown inside footnotes is preserved.
   - Inline code like `@page`, `$x$`, and `[^id]` remains readable when it is written inside backticks.

Final Publishing Checklist

Before publishing an important PDF, check the following items:

- [x] Cover title, subtitle, author, date, and metadata.
- [x] Language and document direction.
- [x] Table of contents depth and labels.
- [x] Pages containing formulas.
- [x] Pages containing code blocks and inline code.
- [x] Pages containing local images.
- [x] Tables that span wide content.
- [x] Footnotes and links.
- [x] Footer numbering after the cover.
- [ ] Final visual review in a PDF viewer.
- [ ] Release checklist completed when publishing a tagged version: `docs/RELEASE.md`.

References

Rastegar, Meraj. (2026a). **Deterministic Publishing for Mixed-Direction Documents**. *Mardas Publishing Notes*. 1, (1), 1–18. doi:10.5555/mardas.2026.1. ↩

Rastegar, Meraj. (2026b). **Offline Citation Pipelines for Reproducible PDF Output**. *Mardas Publishing Notes*. 1, (2), 19–34. ↩

احمدی, مریم. (1404). **انتشار علمی فارسی و مدیریت منابع**. انتشارات پژوهش. ↩